BFI LOAN ORIGINATION · PLATFORM DECISION BRIEF · FOR THE CTO

Two decisions.
Two different deadlines.

DECISION A · THIS MONTH · FIRM

Pay to upgrade Bravo’s workflow engine and Spring Boot. 18–33 engineer-days. No licence needed.

DECISION B · AFTER THREE MEASUREMENTS · OPEN

Do not choose where the next product family is built yet. The evidence does not favour either platform.

How to read this A first, then B, then what we are asking for. Nothing in B holds up A. **As at** 2026-09-11

What Bravo and LORA are, and where the migration has reached

Both platforms originate loans in production today. This is the minimum you need before the two decisions.

BRAVO — THE PLATFORM WE HAVE

- **What it is.** BFI's current loan origination system. Its codebase dates from 2022 and it still runs the retail book.
- **How it works.** The process is *drawn*, as a flowchart in a standard called **BPMN**, and executed by a **Camunda 7** engine inside a single Spring Boot service, `ms-bpm`, behind three operator consoles.
- **What it carries.** **118,253** applications in August — about **41%** of the total, and roughly **95%** of the business value.

LORA — THE PLATFORM WE ARE BUILDING

- **What it is.** The replacement. It has carried the majority of applications since July 2026.
- **How it works.** Nothing is drawn. Go services run on **Temporal**, and a planner works out what can run next from the data a loan already has.
- **What it carries.** **171,479** applications in August — about **59%** of the total, and all of it one product family.

Where the migration has reached. The two run side by side. Bravo still originates the retail book, and **every path in this deck leaves it doing that for at least another 12 months.** So neither decision here is about whether to finish the migration. One is about keeping Bravo safe to run in the meantime. The other is about what the *next* product family gets built on.

TERM	MEANS	TERM	MEANS
BPMN	The flowchart standard Bravo's process is drawn in	GSM	LORA's alternative. Nothing is drawn; the data decides what runs next
Camunda 7	The engine that runs Bravo's flowcharts. We run the free **Community Edition**, which is now out of support	Temporal	The engine LORA runs on, bought as a cloud service
LOS	**Loan Origination System** — everything from application to go-live	ms-bpm	The one Bravo service holding the engine and all the human-task code

Do A this month. Leave B until we have three numbers.

DECISION A · FIRM

This month

How do we make Bravo **safe to run** while we decide B?

Swap the workflow engine and upgrade Spring Boot in one piece of work. It takes 18–33 engineer-days. No licence is needed. It does not wait for B.

DECISION B · OPEN

After three measurements

Where do we build the next product family for the next **5–10 years**?

This one is genuinely open. Cost and business value point to Bravo. Visibility and how often staff must step in point to LORA. It does not wait for A.

> **Whatever we decide, Bravo runs the retail book for at least another 12 months.**
>
> Moving to Temporal would take 12–18 months. Moving to LORA would take 15–24. The Spring Boot upgrade is owed either way. **Nothing later in this deck should delay A.**

FIVE MINUTES

This slide, then **the asks** on slide 13.

FIFTEEN MINUTES

Decision A (4–5), the **value chart** (7), the **measurements** (12), the asks (13).

THE FULL PACK

All 14 slides. Slide 10 covers BPMN against GSM if you want it.

Bravo is already out of support. Someone else's code has already run on its engine.

COMPONENT	BRAVO RUNS	SUPPORT ENDED
Camunda 7 Community Edition	7.23.0	**14 Oct 2025** — the line is closed and the repository is archived. **There will never be another security patch.**
Spring Boot 3.5.x (open source)	3.5.16	**30 Jun 2026**
Java 17 (Oracle premier)	17	30 Sep 2026
Camunda 7.23 Enterprise	**we hold no licence**	13 Oct 2026

Checked in the code: `pom.xml:26` pins 7.23.0. There is no Camunda BOM, no enterprise artifact and no licence.

61 INJECTED PROCESS DEFINITIONS ARE STILL ACTIVE IN PRODUCTION

These are textbook attempts to run code on our server. They were uploaded in 7 sessions between 2026-05-23 and 2026-06-30. **One of them ran.** It read back the name of the machine it ran on. All 61 can still be started.

The upload path `/engine-rest/**` does ask for a credential. So this was **someone using a shared or leaked secret, not an open door**. That shared secret, `INTERNAL_SERVICE_KEY`, gives full control of the engine.

The engine is running today. It gets no security patches, and code can still be uploaded to it.

That is why Decision A is kept separate, and why it does not wait for anything else in this deck.

There is a supported version to move to. It takes 18–33 engineer-days.

Two open-source forks of Camunda 7 are available. Both run on Spring Boot 4. Both keep our existing `ACT_` database tables and accept all 53 of our current BPMN files. Java 17 stays as it is. By comparison, rewriting on Temporal or moving to LORA would take **31–57 engineer-months.**

	OPERATON 2.1.4	CIB SEVEN 2.2.0
Commits in the last year	**2,310**	258
People contributing	**52**	31
Share written by one person	~80%	**25%**
Former Camunda engineers	none	**two**
Can we buy support?	**no** — each release lasts 6 months	**yes**, but terms are not published
Needs an interim hop to 7.24 first	**yes**	**no**

THE ONE QUESTION LEFT TO ANSWER

Do we need a support contract we can actually buy?

Yes → choose CIB seven. No → choose Operaton.

IF THE ANSWER MIGHT BE YES, ASK CIB NOW

CIB does not publish contract length, service levels or price. You have to ask them directly. Start that conversation before we choose, not after.

RUN FOUR DATABASE CHECKS BEFORE SETTING A CUTOVER DATE

One table, `ACT_GE_SCHEMA_LOG`, may still hold a version label from before 7.22. If it does, the new engine will not start. Checking takes a few hours. Do it first.

B is only about where the next product family is built.

YOU ARE DECIDING THIS

— **Whether you can see what is happening.** Bravo's workflow layer produces no tracing data at all. It has 38 monitors, and none of them watch a loan's progress.

— **Whether you can test it.** Bravo's one scheduled end-to-end test cannot report a failure. The tests behind it have not changed since November 2023.

— **How much this grows.** Every new family built on Bravo adds more of both problems.

— **Who you can hire.** Java, Spring and Camunda are common skills. Go plus our own custom planner is not. This is the one staffing point you cannot fix by moving people around.

YOU ARE NOT DECIDING THIS

— **BPMN against GSM.** Slide 10 covers it. The choice shaped about 6% of Bravo and 10% of LORA. It is real, it is settled, and it does not decide B.

— **Whether Bravo works.** It does. It handled 118,253 applications in August. When things fail, they fail cleanly, and no loan gets stuck forever.

— **Whether to retire Bravo faster.** We do not recommend it.

— **Which platform is cheaper.** Bravo is, by 4–5 times per application. Settled.

— **Team sizes.** That is something you control. It is not a fact about either platform.

— **The engine upgrade.** That is Decision A, and it already has an answer.

Judge a modelling approach by what teams actually build under deadline pressure, not by what it allows in theory.

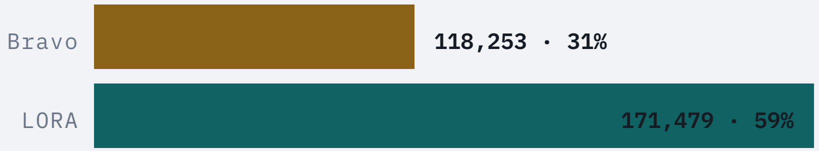
We accept the Bravo team's review. The flowchart problems come from how Bravo was modelled, and standard fixes exist. Of those three fixes, one is **5.5% rolled out**, one has **not been started**, and one is **not enforced**.

06 / DECISION B — THE PICTURE THAT CHANGED

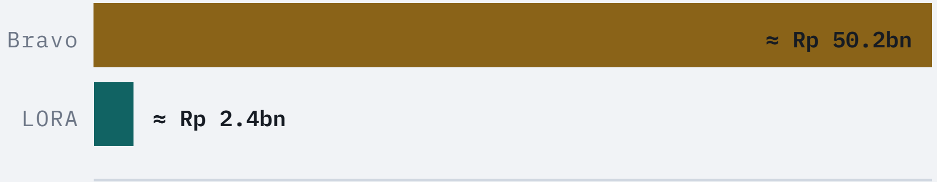
LORA handles more applications. Bravo brings in the money.

Every rate in this pack is worked out per application, using counts from a billing sheet. On that measure LORA has 59% and Bravo has 41%. **Measured by business value, it is the other way round.**

APPLICATIONS, AUG 2026



BUSINESS VALUE



VALUE PER APPLICATION

Bravo ≈ Rp 425,000 **LORA ≈ Rp 14,000**

That is about 30 times more value per application on Bravo.

ALL OF LORA'S 59% IS ONE PRODUCT FAMILY

Only NDF runs in production. NDF2W is 90.2% of it and NDF4W is 9.8%. **Six of the seven families are not deployed.** Three of those have no production deploy pipeline at all. There is no switch to flip.

BRAVO HAS FOUR FAMILIES LIVE AND A FIFTH IN UAT

NDF2W, NDF4W, RO and DF4W are in production. **DF2W is in UAT**, with a live security test and open go-live tickets. Families that were never launched tell you nothing about how easy a platform is to build on. That is true of both sides.

THE TWO RUPIAH FIGURES ARE AN ESTIMATE FROM THE CTO OFFICE

Neither platform holds agreement, disbursement, new-money or revenue data. Getting this split out of a real system, with the period and the basis written down, is **the first of the three measurements B waits for.**

Who is ahead on what

BRAVO

— **5 to 6.5 times cheaper per application**
Rp 490–515 each against Rp 2,500–3,200. The workflow tier costs Rp 58M a month against Rp 431M.

— **About 21 times the business value**
Rp 50.2bn against Rp 2.4bn. About 30 times more per application.

— **Four live product families, not one**
Plus DF2W in UAT. LORA's other six have never taken an application.

— **A change touches one or two repositories**
LORA needs five, in order: schema, handler, activity, permissions, front end.

— **Easy to hire for**
Java, Spring and Camunda. LORA needs Go plus a planner you cannot hire for.

— **You can ask questions across all loans**
"Which loans are stuck in survey?" is one query. LORA cannot answer it.

— **Failures are contained and labelled**
Nothing gets stuck forever. Failed loans wait in an operator console.

— **You can tell which upstream is failing**
Bravo keeps 25 of them separate. LORA puts about 300 behind one route.

LORA

— **Staff step in 3.1 times less often**
0.126% of applications against 0.389% — 1 in 790 against 1 in 257. One ticket type accounts for the gap, and it was fixed on 2026-09-10.

— **Every step is traced, with the loan id**
Bravo produces no tracing at all for its workflow layer.

— **A failure points to a customer**
The workflow id *is* the application. Bravo's logs carry no application id.

— **Extra volume is almost free**
Volume rose 23% for 3% more spend. Bravo's unit cost rises as volume moves away.

— **Deploys leave running loans alone**
Each version has its own queue. On Bravo, a shared sub-process redeploy reaches 73% of volume.

— **An end-to-end test that really checks**
Mostly red at night, which at least means it works. Bravo's run cannot fail, and has not changed since 2023.

— **Tests can force an upstream to fail**
Two lines of code. Bravo has never run its 50 error handlers or 190 escalations.

— **Products are kept apart by design**
Separate documents and separate queues. Proven on one family only.

Left off on purpose. Team size, 25 against 22, is your decision, not a platform fact. Ticket volume flipped once Bravo's full scope was counted. Cycle time is 2 days on both. Neither side enforces coverage. Full table: `compare.md` §1.

Three facts decide B. One measurement could flip two of them.

4–5×	3.5×	0	3
CHEAPER PER APPLICATION ON BRAVO, WHICH ALSO CARRIES ABOUT 21 TIMES THE VALUE	MORE OFTEN BRAVO NEEDS A PERSON. ONE TICKET TYPE COULD CLOSE THE GAP	TRACES, LOAN-LEVEL MONITORS AND WORKING END-TO-END TESTS ON BRAVO'S WORKFLOW LAYER	MEASUREMENTS THAT WOULD SETTLE THIS. NONE HAS BEEN TAKEN

1 · COST AND VALUE — POINTS TO BRAVO

Bravo's workflow tier costs about Rp 58M a month. LORA's costs about Rp 431M. Per application Bravo is 4–5 times cheaper, and it carries about 21 times the business value.

Two things limit the argument. LORA's gap comes from contracts and over-provisioning, not from how it is built. And Bravo's advantage shrinks as volume moves away: at 40,000 a month it would be about Rp 1,450 each.

Separately, Cloud Logging alone costs Rp 140.5M a month. That is 2.4 times the tier a migration would remove, and it is fixed by changing configuration.

2 · HOW OFTEN STAFF STEP IN — POINTS TO LORA, BUT BARELY

Bravo needs a person on 0.389% of applications. LORA needs one on 0.126%.

Almost all of that gap is one ticket type, `Surveyor Platform - Release reject`. It is 28.4% of Bravo's tickets and has grown 4.8 times. Nobody has traced it to a code path yet.

Take it out and Bravo is at 0.134% against LORA's 0.126%. Tracing it is hours of work. **This is the second measurement B waits for.**

3 · VISIBILITY AND TESTS — POINTS TO LORA, AND IS CHEAP TO FIX

Bravo's workflow layer produces no traces. Its logs carry no loan id. None of its monitors watch a loan's progress. Its end-to-end suite is frozen, so it cannot fail.

The two kinds of test that would catch the most run in milliseconds and are about 7 days of work. **This is a decision nobody has made, not a mountain to climb.** Fund the fix. It is a weak reason to move products.

TWO THINGS THAT LOOK LIKE REASONS AND ARE NOT

Team size is the result of the last decision to consolidate on LORA. Quoting it back to the office that made that decision proves nothing. **"Seven product families"** was never true in production, because only one family is live. Both are off the board.

BPMN against GSM is settled. It is not what decides B.

THE VERDICT ON THE ORIGINAL RATIONALE

The choice of modelling approach shaped about 6% of Bravo and 10% of LORA. That part is the automated pipeline, and LORA is better there. The other ~90% looks much the same on both. What differs in that 90% is engineering discipline.

LORA'S RATIONALE CLAIMED	WHAT THE CODE SHOWED	VERDICT
Flowcharts grow out of control with every edge case	True here. `ndf4w.bpmn` has 111 service tasks, 137 gateways and 9,494 lines.	● LORA
Loan events happen in parallel, and flowcharts handle that badly	Bravo is sequential by choice. Camunda supports what is missing. Nobody drew it.	● LORA
Workflow data ends up scattered across process variables	Not how Bravo did it. Business data sits in 238 JPA entities. What is scattered is lifecycle state.	● REFUTED
Temporal retries automatically; engines need hand-built plumbing	Bravo built that plumbing and ended up with a better policy: bounded retries, a degraded last attempt, an operator queue.	● BRAVO
Undoing a failed step is the execution layer's job	Bravo has no compensation events. LORA only rolls back document fields. Both fall back on sweepers and people.	● NEITHER

THE BRAVO TEAM REVIEWED THIS, AND WE ACCEPTED THEIR REVIEW

The flowchart problems and the sequential flows are how Bravo was modelled. They are not laws of BPMN, and standard fixes exist. Of those fixes, one is 5.5% rolled out, one has not been started, and one is not enforced.

Seven further things are the same on both platforms: human-task complexity, untested orchestration, an abandoned end-to-end suite, an unenforced lifecycle, no compensation logic, the same misleading health model, and surveyor assignment as the top complaint. The full write-up is in `compare-architecture.md`.

What you accept, either way

IF THE NEXT PRODUCT GOES TO BRAVO

— **You take on a community fork as a dependency.** Operaton supports each release for 6 months. CIB seven sells a contract, but will not publish its terms.

— **The new product will be as hard to see into as the current ones,** until traces and monitors are funded. Failures will reach you as tickets.

— **Redeploying a shared sub-process affects loans already in flight.** That is 73% of volume today, and there is no migration plan.

— **You inherit five separate ways of telling products apart.** Adding a product touches all five, and no test shows which products a change hits.

— **You get the cheaper platform** and the lowest cost per change in the estate. That is the trade.

IF THE NEXT PRODUCT GOES TO LORA

— **Every integration change costs five repositories.** This is real and measured. A generator would fix it, but it has only been designed, not built.

— **You inherit retries that never give up.** About half of loans never finish, while every signal says the system is healthy.

— **You cannot ask questions across all loans,** and you cannot tell which upstream is degrading until a merged change is released.

— **You are betting on one proven family.** Six of the seven have never been deployed, and conversion is unmeasured: about Rp 14,000 per application against Bravo's Rp 425,000.

— **You get near-zero cost per extra application,** products kept apart, deploys that leave running loans alone, and a test that actually checks. That is the trade.

LORA's risk is that it has not been proven at scale. Bravo's risk is discipline that has not been fixed. Both can be dealt with, and both have already been scoped. That is why B is still open.

Three measurements gate B. None has been taken.

#	MEASURE THIS	IF IT COMES BACK ONE WAY	IF IT COMES BACK THE OTHER	COST
1	**The agreement, new-money and disbursement split by platform.** From a real system, with the period and the basis written down.	It confirms the 21-times gap. **B tilts to Bravo** for anything that earns revenue, and LORA's volume argument goes away.	Conversion is similar on both. **B tilts back to LORA.**	**days**
2	**Confirm the `Surveyor Platform - Release reject` fix.** Fixed and deployed 2026-09-10, squad `LN`.	It is a fixable defect. Bravo's rate falls to about 0.134% against 0.126%, and **LORA's reliability lead goes away.**	It is a built-in cost of how Bravo works. The 3.5-times gap stands.	**hours**
3	**What it costs to launch one undeployed LORA family.** Three of the six have no production deploy pipeline.	It is cheap. "Put the next family on LORA" becomes a real proposal.	It is expensive. The proposal has been assumed rather than priced. Nobody has launched a family there since NDF.	weeks

THE ONE ITEM HERE THAT SHOULD NOT WAIT FOR EVIDENCE

Fund Decision A. Once the engine and Spring Boot are done, B can be decided on architecture and economics alone. It costs 18–33 engineer-days.

Take A now. Defer B. Approve four things.

DECISION A · TAKE IT NOW	DECISION B · DEFER
Fund the engine and Spring Boot upgrade. Close the security hole this week. **18–33 engineer-days. No licence needed.**	Do not place the next family until measurements 1 and 2 are in. They cost **days**. Either one could decide it.

1	**Close the Camunda security hole this week.** Rotate `INTERNAL_SERVICE_KEY`. Find out who uploaded the code, from the deploy logs between 2026-05-23 and 2026-06-30. Turn on engine authentication. Delete the 61 definitions.	THIS WEEK EVERY OPTION
2	**Fund the engine and Spring Boot upgrade.** First answer one question: do we need a support contract we can actually buy? Then run the database checks before setting a cutover date.	18–33 ENG-DAYS RP 45–125M
3	**Commission the agreement and new-money split by platform,** with the period and basis written down. Define what counts as a Bravo application and a LORA application. The 47-times ratio is still only an estimate.	DAYS GATES B
4	**Approve the Cloud Logging fix.** It saves Rp 50–90M a month and is configuration only. That is more than retiring Bravo's workflow tier would save. Do it whatever you decide about B.	WEEKS RP 50–90M / MO

ONE STANDING INSTRUCTION

Bravo carries most of the book and most of the money. Resource it as a strategic platform, not as a system being wound down.

Where the numbers come from, and what is still weak

MEASURED FOR THIS ANALYSIS

Jira — projects `DF`, `D2W`, `BL`, `BLCS` and `LN`.

Datadog — tracing, 1.1M log lines a week, the full monitor list, Synthetics, CI Visibility, and browser data from four consoles.

GCP billing through the FinOps API, for the whole of August 2026.

Both codebases — `bravo-bpm-service` at v2.93.43, three Bravo consoles, `bravo-e2e-test`, and 13 LORA repositories.

OTRS ticket export for both platforms, over the same window.

KNOWN WEAKNESSES

The application counts. They come from a billing sheet that nothing else confirms. In every per-application figure the top number is verified and the bottom number is disputed. Ask 3 fixes this.

Bravo keeps the hard cases. As work moves off it, what is left is the difficult residue. We have not corrected for that.

One ticket type carries the second fact. Six named endpoints, squad `LN`. That is hours of instrumentation, not an open-ended search.

The Rp 50.2bn and Rp 2.4bn figures are an estimate from the CTO office. The period and basis still need confirming.

READ FURTHER

`compare.md` — the 40-row comparison, with a source on every row, and the recommendation in full.

`compare-architecture.md` — BPMN against GSM in detail, plus the Bravo team's review.

Five findings documents — people, testing, observability, cost and delivery. Each has its own deck.

`ticket-analysis.md` — how often each platform needs a person.

`option-summary.md` — Decision A in full, including the fork comparison.

Decision A has a firm answer and costs 18–33 engineer-days. Decision B is open, and the two measurements that would settle it cost days.

Four earlier readings in this pack turned out to be wrong. Bravo's end-to-end suite does exist, but is frozen. Team size was used as evidence, which is circular. "Seven LORA families" was really one. And Bravo's consoles and the `LN` project were never counted. **Every one of those corrections moved in Bravo's favour.** The findings that survived — the dead engine, the blind workflow layer, the cost ratio and the intervention gap — are the ones to trust.